

Creation Date 22-Sep-2009

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Revision Number 1

1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/UNDERTAKING

Product Name GRAM ENHANCER
Cat No. R40231, R40232
Synonyms No information available.
Recommended Use Laboratory chemicals

Company Remel
12076 Santa Fe Drive
Lenexa, KS 66215 United States
Telephone: 1-800-255-6730
Fax: 1-800-621-8251

Emergency Telephone Number INFOTRAC 1-352-323-3500 (International)
1-800-535-5053 (North America)

2. HAZARDS IDENTIFICATION

Classification according to European directive on classification of hazardous preparations 1999/45/EC.

Xn



R -phrase(s)

none
R40 - Limited evidence of a carcinogenic effect
R43 - May cause sensitization by skin contact

3. COMPOSITION/INFORMATION ON INGREDIENTS

Haz/Non-haz

Component	Weight %	EC No.	Classification
Formaldehyde 50-00-0	1 - 4	-	Carc.Cat.3;R40 T;R23/24/25 C;R34 R43
Methyl alcohol 67-56-1	0.25 - 2	EEC No. 200-659-6	F;R11 T;R23/24/25-39/23/24/25

For the full text of the R phrases mentioned in this Section, see Section 16

4. FIRST AID MEASURES

Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Obtain medical attention.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.
Ingestion	Clean mouth with water. Obtain medical attention.
Inhalation	Move to fresh air. If breathing is difficult, give oxygen. Get medical attention immediately if symptoms occur.
Notes to Physician	Treat symptomatically.

5. FIRE-FIGHTING MEASURES

Suitable Extinguishing Media

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

Extinguishing media which must not be used for safety reasons

No information available.

Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition

Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

Flash Point	No information available.
Method	No information available.
Autoignition Temperature	No information available.

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions	Ensure adequate ventilation. Use personal protective equipment.
Environmental Precautions	Should not be released into the environment.
Methods for Containment and Clean Up	Soak up with inert absorbent material Keep in suitable and closed containers for disposal

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7. HANDLING AND STORAGE

Handling	Wear personal protective equipment. Ensure adequate ventilation. Avoid contact with skin, eyes and clothing. Avoid ingestion and inhalation.
Storage	Keep containers tightly closed in a dry, cool and well-ventilated place.
Specific use(s)	No information available.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Exposure limits

Component	European Union	The United Kingdom	France	Spain	Germany
Formaldehyde			VME: 0.5 ppm VLCT: 1 ppm	VLA-EC: 0.37 mg/m ³ VLA-EC: 0.3 ppm	
Methyl alcohol	TWA: 200 ppm TWA: 260 mg/m ³ Skin		VME: 200 ppm VME: 260 mg/m ³ VLCT: 1000 ppm VLCT: 1300 mg/m ³	VLA-ED: 266 mg/m ³ VLA-ED: 200 ppm	

Component	Italy	Portugal	The Netherlands	Finland	Austria
Formaldehyde		Ceiling: 0.3 ppm	STEL: 0.5 mg/m ³ TWA: 0.15 mg/m ³	TWA: 0.37 mg/m ³ TWA: 0.3 ppm STEL: 1.2 mg/m ³ STEL: 1 ppm Ceiling: 1 ppm Ceiling: 1.2 mg/m ³	STEL: 0.5 ppm STEL: 0.6 mg/m ³ MAK: 0.5 ppm MAK: 0.6 mg/m ³ Ceiling: 0.5 ppm Ceiling: 0.6 mg/m ³
Methyl alcohol	TWA: 200 ppm TWA: 260 mg/m ³	STEL: 250 ppm TWA: 200 ppm	STEL: 520 mg/m ³ TWA: 260 mg/m ³	TWA: 270 mg/m ³ TWA: 200 ppm STEL: 330 mg/m ³ STEL: 250 ppm	STEL: 1040 mg/m ³ STEL: 800 ppm MAK: 260 mg/m ³ MAK: 200 ppm

Component	Switzerland	Poland	Norway	Ireland	Denmark	Australia
Formaldehyde	STEL: 0.6 ppm STEL: 0.74 mg/m ³ MAK: 0.3 ppm MAK: 0.37 mg/m ³	NDSch: 1 mg/m ³ NDS: 0.5 mg/m ³	TWA: 0.5 ppm TWA: 0.6 mg/m ³ Ceiling: 1 ppm Ceiling: 1.2 mg/m ³	TWA: 2.5 mg/m ³ TWA: 2 ppm STEL: 2.5 mg/m ³ STEL: 2 ppm	Ceiling: 0.3 ppm Ceiling: 0.4 mg/m ³	STEL: 2 ppm STEL: 2.5 mg/m ³ TWA: 1.2 mg/m ³ TWA: 1 ppm

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Component	Switzerland	Poland	Norway	Ireland	Denmark	Australia
Methyl alcohol	STEL: 1040 mg/m ³ STEL: 800 ppm MAK: 200 ppm MAK: 260 mg/m ³	NDSch: 300 mg/m ³ NDS: 100 mg/m ³	TWA: 100 ppm TWA: 130 mg/m ³	TWA: 200 ppm TWA: 260 mg/m ³ STEL: 250 ppm STEL: 310 mg/m ³ Skin	TWA: 200 ppm TWA: 260 mg/m ³	STEL: 250 ppm STEL: 328 mg/m ³ TWA: 262 mg/m ³ TWA: 200 ppm

Occupational exposure controls
Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location Ensure adequate ventilation, especially in confined areas

Personal Protective Equipment
Respiratory Protection

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Eye Protection

Tightly fitting safety goggles

Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Hand Protection

Protective gloves

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

Environmental exposure controls

No information available.

9. PHYSICAL AND CHEMICAL PROPERTIES
Physical State

Liquid

Boiling Point/Range

No information available.

Melting Point/Range

No information available.

Flash Point

No information available.

10. STABILITY AND REACTIVITY
Stability

Stable under normal conditions.

Conditions to Avoid

Incompatible products. Excess heat.

Incompatible Materials

Strong oxidizing agents

Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO₂)

Hazardous Polymerization

Hazardous polymerization does not occur.

Hazardous Reactions .

None under normal processing..

11. TOXICOLOGICAL INFORMATION
Acute Toxicity

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Component Information

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Formaldehyde	500 mg/kg (Rat)		0.578 mg/L (Rat) 4 h
Methyl alcohol	5628 mg/kg (Rat)	15800 mg/kg (Rabbit)	64000 ppm (Rat) 4 h 83.2 mg/L (Rat) 4 h

Chronic Toxicity
Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	IARC	UK
Formaldehyde	Group 1	

IARC: (International Agency for Research on Cancer)

IARC: (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

Sensitization

No information available.

Neurological Effects

No information available.

Mutagenic Effects

No information available.

Reproductive Effects

No information available.

Developmental Effects

No information available.

Target Organs

None known., Gastrointestinal tract (GI), Central nervous system (CNS), Eyes, Respiratory system, Skin, Optic nerve.

12. ECOLOGICAL INFORMATION
Ecotoxicity

Component	Freshwater Algae	Freshwater Fish	Microtox	Water Flea
Formaldehyde				EC50 96 h 20 mg/L EC50 48 h 2 mg/L EC50 96 h 20 mg/L
Methyl alcohol			EC50 = 39000 mg/L 25 min EC50 = 40000 mg/L 15 min EC50 = 43000 mg/L 5 min	

Persistence and Degradability

No information available

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Bioaccumulative Potential No information available.

Mobility No information available.

Component	log Pow
Formaldehyde	0.35
Methyl alcohol	-0.77

13. DISPOSAL CONSIDERATIONS

Waste from Residues / Unused Products Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification

Contaminated Packaging Dispose of in accordance with local regulations.

14. TRANSPORT INFORMATION

IMDG/IMO Not regulated

ADR Not regulated

IATA Not regulated

15. REGULATORY INFORMATION

Classification according to European directive on classification of hazardous preparations 1999/45/EC

Labelling

Xn



R -phrase(s)

none

R40 - Limited evidence of a carcinogenic effect

R43 - May cause sensitization by skin contact

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S -phrase(s)

S24/25 - Avoid contact with skin and eyes

International Inventories

Component	EINECS	ELINCS	NLP	TSCA	DSL	NDSL	PICCS	ENCS	CHINA	AICS	KECL
Formaldehyde	200-001-8	-		X	X	-	X	X	X	X	KE-17074 X
Methyl alcohol	200-659-6	-		X	X	-	X	X	X	X	KE-23193 X

16. OTHER INFORMATION

Text of R phrases mentioned in Section 2-3

R11 - Highly flammable

R34 - Causes burns

R40 - Limited evidence of a carcinogenic effect

R43 - May cause sensitization by skin contact

R23/24/25 - Toxic by inhalation, in contact with skin and if swallowed

R39/23/24/25 - Toxic: danger of very serious irreversible effects through inhalation, in contact with skin and if swallowed

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Literary reference	No information available.

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End of Safety Data Sheet